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Palaeomagnetism of the Montagnais Diabase Dikes of the West Central Sector of the Labrador Trough

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Summary – Eighty-six specimens were collected from eighteen different sites on the numerous north-striking thin diabase dikes of middle-Proterozoic (Helikian) age in the west central portion of the Labrador trough. These diabase dikes, which are the only intrusive rocks found in the area investigated (longitude 66°00′–67°30′W, latitude 54°00′–55°30′N), consist mainly of plagioclase feldspars, olivine, pyroxenes and magnetite. Magnetite is the magnetic memory carrier, its Curie point (580°C) was determined with an electromagnetic balance. The volume ferromagnetic susceptibility, the intensity and orientation of natural remanent magnetization (N.R.M.) were measured in the presence of DC magnetic fields of 0.5 to 1 oersted. In order to obtain information relative to the stability of the N.R.M. component, stepwise alternating field (AF) demagnetization was conducted at intensities of 50, 100, 150, 200, 250, 300, 400 and 500 oersted (peak-to-peak) on twenty-one specimens, that is, roughly 25% of the total number. A palaeomagnetic stability index test allowing one to choose the AF intensity range for which the rate of change in remanence direction of a diabase specimen is minimized was used. The optimum AF intensity, at which the primary thermoremanent component is most thoroughly isolated after most of the isothermal remanent magnetization (IRM) has been removed and before significant anhysteric remanent magnetization (ARM) is introduced, is located in the 100–200 oersted range. The remaining 75% of the specimens was then demagnetized at steps of 100, 150 and 200 oersteds. The decrease in intensity of the N.R.M. component in the course of AF demagnetization is quite rapid between 0.5 and 150 oersteds and relatively insignificant between 150 and 500 oersteds or more. A mean paleomagnetic pole position of 137°W, 10°S ($\alpha\,0.95 = 14°$, $d_p = 08°$, $d_m = 15°$, $K = 17$), with at least one reversal, indicates that the Labrador trough was near the equator (Central North Pacific Ocean) at the time of intrusion. The possible implications of the results of this study are discussed in relation to hypotheses on polar wandering and on continental drift.

1. Introduction

Diabase dike swarms are numerous across the Canadian Shield and have been observed in all structural provinces. The large arrays of basic intrusive and hypabyssal rocks produced by important episodes of igneous activity in Precambrian time are characterized by internal uniformity of K–Ar ages and by uniformity of palaeomagnetic directions (LAROCHELLE, 1967; FAHRIG and JONES, 1969; FAHRIG, IRVING and JACKSON, 1971; MURTHY, 1971). The dikes are well distributed in time and space and are very informative regarding the tectonic development of the Canadian Shield (IRVING, DONALDSON and PARK, 1972; FAHRIG and SCHWARZ, 1973). Of particular interest are the diabase dikes in the Churchill structural province within the central sector of the

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Labrador trough. The Montagnais diabase dikes are not numerous and are often poorly exposed. They derive their name from the Montagnais Indian tribe living in the vicinity of Schefferville.

This study concerns the dikes, both petrographically and palaeomagnetically, and new data are presented from both aspects. The objects of this paper are two-fold, firstly to obtain chemical, petrographic and palaeomagnetic data from the Montagnais dikes and, secondly, to derive a pole position for them from stability tested data. Since palaeomagnetism takes a greater importance in determining the time order of Precambrian rocks and in studying the nature of Precambrian orogenesis, many detailed palaeomagnetic studies ought to be conducted in order to determine the variations of the Precambrian geomagnetic field. This work is a small contribution to this general requirement, but it is of some interest since no definite trend of apparent polar wandering has yet been established for the 1000–1500 m.y. span. These uncertainties call for additional results from rocks that were magnetized in that period and in this respect the Labrador trough is an untapped source of new palaeomagnetic data.

2. Geology

The diabase dikes of the central part of the Labrador trough belong to a system of olivine dolerite sheets intruded into the Lower Proterozoic (Aphebian) sedimentary and volcanic rocks of the trough in northern Quebec and Newfoundland-Labrador (Fig. 1). The sector was geologically mapped on a regional scale by the Geological Survey of Canada (FRAREY, 1961, 1967) and in detail by the Labrador Mining and Exploration Co. and the Iron Ore Company of Canada at different intervals of time since 1946. The stratigraphical sequence of the sedimentary and volcanic rocks of the trough is presented in HARRISON (1952), GASTIL *et al.* (1960), FRAREY and DUFFELL (1964), and DIMROTH *et al.* (1970). The sequence includes, from bottom to top, slates, dolomite, chert breccia, quartzite, iron formation members, metabasalts and pyroclastics. The thickness of the whole sequence amounts to 10 000 m or more.

The dikes, which are rarely well-exposed, cut the strike of the Kaniapiskau system of Lower Proterozoic rocks at an angle of 45° and run approximately north–south. They are vertical and vary in thickness from 5 to 70 m, the average size being some 13 m in width. The dikes are relatively short and seldom exceed a few tenths of kilometers in length.

3. Petrology and chemistry

The dikes are composed of fresh to occasionally altered olivine dolerite. The diabases are composed primarily of plagioclase, pyroxene, olivine and magnetite. They are medium- to coarse-grained and have ophitic, sub-ophitic and intergranular textures. The composition of plagioclase varies from 48 to 70% anorthite. The plagioclase crystals are variously twinned, and zoning is present in untwinned crystals. The pyroxenes include augite, subcalcic augite and pigeonite (hypersthene and ferro-augite or

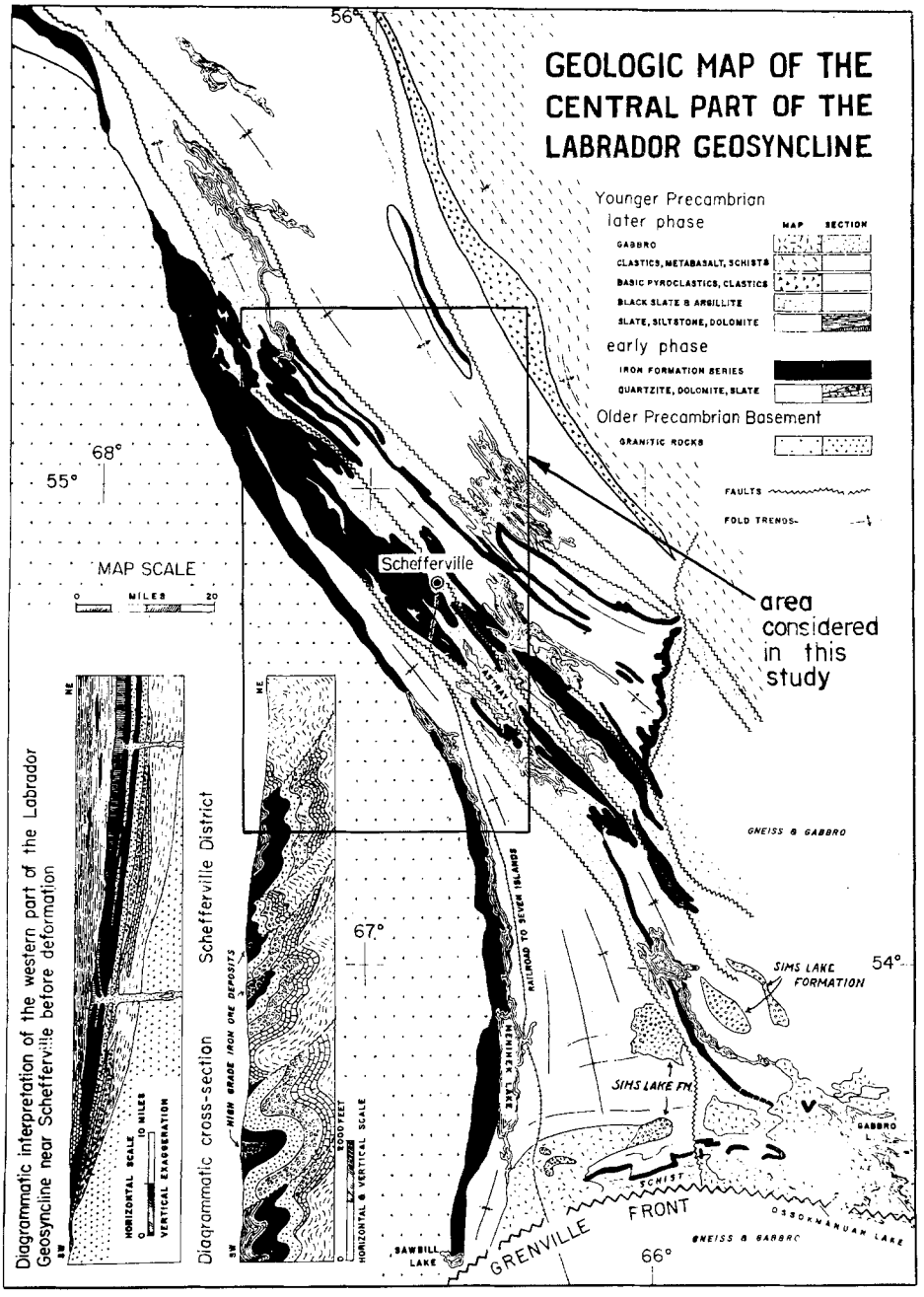


Figure 1
Geological map of the central part of the Labrador geosyncline (after Gastil, Blais, Knowles and Bergeron). The area investigated is outlined by the central rectangle

hedenbergite are very rare), augite being the most abundant mafic mineral. Some pigeonite is zoned and a few grains are uniaxial. Development of pyroxene overgrowth is also observed. Olivine occurs as a subordinate mineral in relation to plagioclase and pyroxenes. In places, it shows ophitic texture with plagioclase. The composition of olivine varies from $Fo_{42-46}Fa_{58-54}$ to $Fo_{65-78}Fa_{35-22}$. The alteration products which represent a very small fraction of the volume are derived from the pyroxenes and olivine. They consist of chlorite, bowlingite, serpentinite, iron oxides (geothite and hematite) and carbonates. Magnetite is believed to be primary.

The mean of seven chemical analyses performed on these olivine diabases yielded the following results: 46.45% SiO_2, 2.20% TiO_2, 15.76% Al_2O_3, 2.65% Fe_2O_3, 12.59% FeO, 0.22% MnO, 6.20% MgO, 8.20% CaO, 2.86% Na_2O, 0.90% K_2O, 0.36% P_2O_5, 1.27% H_2O^+, 0.03% H_2O^-, 0.18% CO_2 and 0.13% S. The ferrous/ferric ratio is quite high for the rocks as a whole, indicating that they are in a quite reduced state and that the oxidation state of the opaque minerals is low.

4. Age

The relative age of intrusion of these dikes cannot be established with precision. The diabases are definitely post-Aphebian (<1946 m.y.) as they intrude the Kaniapiskau system and definitely pre-Cretaceous (>135 m.y.) as some of these dikes were affected by the short-lived or low-intensity orogeny which occurred in the area during the Cretaceous. Two whole rock, K–Ar ages were obtained on these dikes and they indicate ages of 1255 ± 52 and 1146 ± 61 m.y., that is Neo-Helikian. According to Dr. W. Fahrig (written communication), the value of 1145 m.y. is more reliable as it was obtained on chilled material and was analyzed later in the history of the radiometric age-dating program of the Geological Survey of Canada when equipment had been much improved (Dr. R. K. Wanless, personal communication). The degree of alteration is definitely too small to alter the radiogenic ages.

5. Sampling procedure

From 1968 to 1972, forty oriented samples were collected by the hand-sample method from eighteen different sites (SEGUIN, 1968, 1969, 1972b); the number of samples per site varied between two and five. Two specimens (occasionally three) were cut for each hand-sample. In this regional survey, efforts were made to select sites where the dikes were well exposed in order to obtain a complete set of samples and to know the geometry of the dikes; great care was taken to select fresh material. The samples locations were chosen by reference to detailed geological maps obtained from the Labrador Mining and Exploration Co. and the Iron Ore Company of Canada. In many places geological maps with a scale as accurate as 200 feet to the inch were used to position the locations. Because of the extensive occurrences of iron formations in the region studied, a magnetic compass could not be used for the orientation of the

rock samples; instead a solar compass and standard surveying instruments were used. The hand-samples were either cubed or cored in the laboratory with an accuracy in the orientation of the order of $\pm 2°$.

6. *Measurements*

Magnetic susceptibility measurements were made with a standardized resistance bridge manufactured by Cenco Instruments Corporation, having a precision of 5×10^{-5} c.g.s. e.m.u. (SEGUIN, 1972a, c). The direction and intensity of remanent magnetization of each specimen was measured by means of a custom-built spinner magnetometer. Alternating field demagnetization was carried out to remove unwanted secondary components using the demagnetizer, built at Université Laval, having a maximum peak-to-peak intensity of 2000 oersteds.

7. *Alternating field demagnetization*

At least one specimen from each of the eighteen sites was demagnetized, in steps of 50 oersteds, from 50 to 550 oersteds and occasionally up to 1000 oersteds in the absence of an ambient field. Guided by the step-wise changes in remanence and intensity of

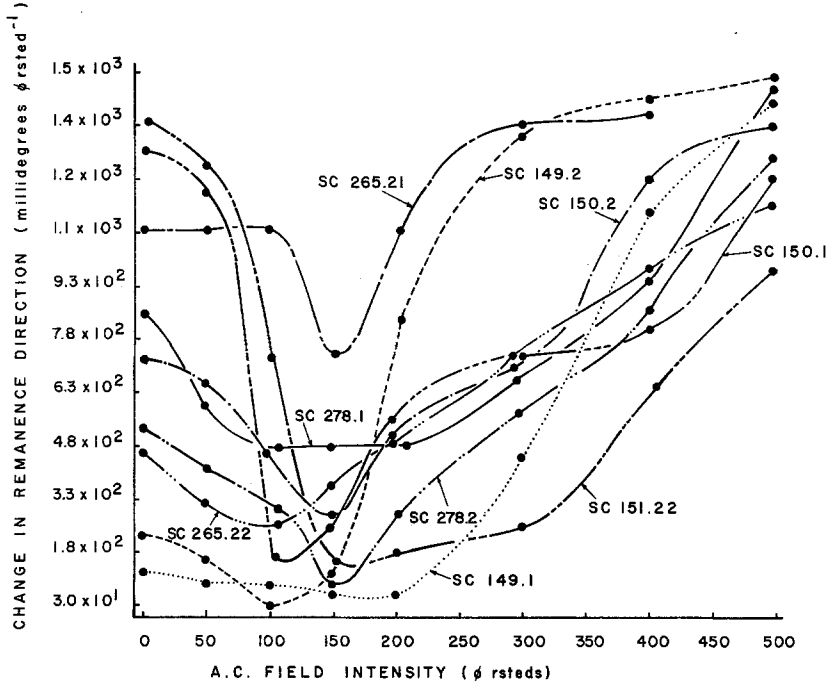


Figure 2
Plot of the stability index of some Montagnais diabase dikes specimens as a function of the A.C. demagnetization field strength

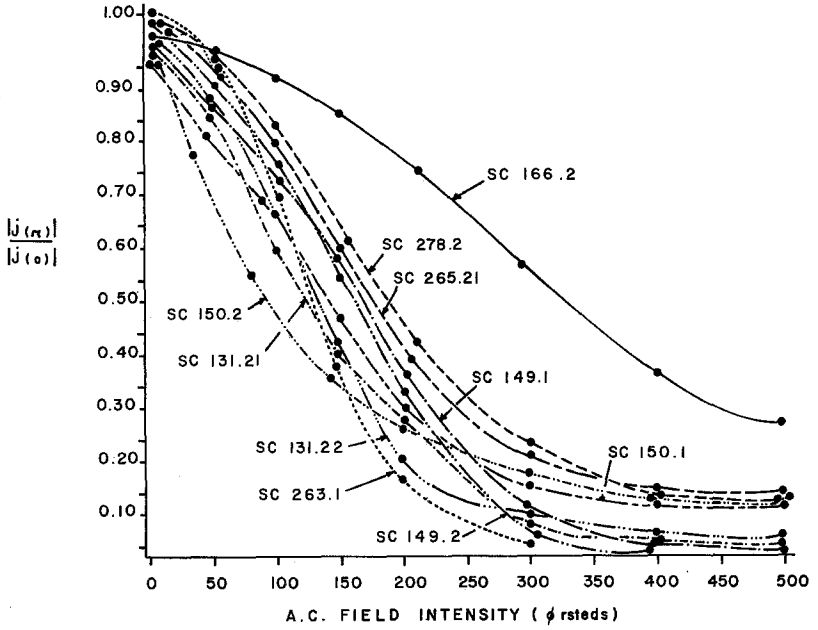


Figure 3
Plot of normalized remanent intensity of some Montagnais diabase dikes specimens as a function of the demagnetizing field strength

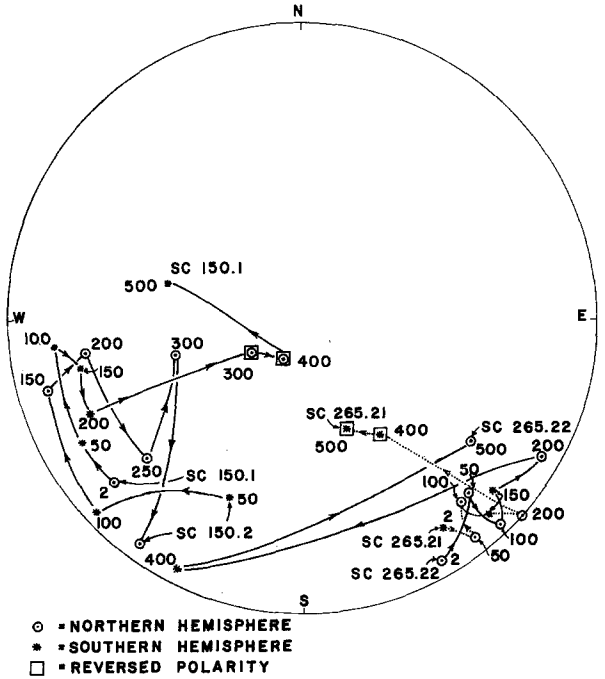


Figure 4
Variation of the magnetization directions of a few pilot specimens of Montagnais diabases as a function of the demagnetizing AF strength

the pilot specimen(s) of the site an intensity of AF demagnetization was selected using an objective 'end point' stability method (SYMONS and STUPAVSKY, 1973) in which the remaining specimens from the site were AF demagnetized and their residual remanence remeasured. The pilot test has shown that the AF intensity at which the primary thermoremanent component is most thoroughly isolated after most of the IRM has been removed and before significant ARM is introduced is located in the 100–200 oersted range (Fig. 2). A normalized demagnetization curve for a test specimen from a few sites is given in Fig. 3. The magnetization directions of pilot specimens from the same sites are presented in Fig. 4. All the other specimens were AF demagnetized at successive intensities of 100, 150 and 200 oersteds. The entire study involved approximately 500 measurements of remanence.

8. *Statistics and grouping*

The vector sum of the unit vectors, representing the magnetization of the specimens from a given hand-sample, was calculated and taken as the mean direction. If the vectors of magnetization diverged by an angle exceeding 23° they were termed 'inhomogeneous' and were not included in any further analysis. This procedure, which was originally suggested by LAROCHELLE (1966), is based on a minimum within-sample precision of 25 ($n = 2$). Instead of applying this test before AF demagnetization, as was done by Larochelle, it was applied after demagnetization. The magnetization of 74% of the

Table 1
Mean directions of remanent magnetization and pole position after magnetic cleaning at 100, 150 and 200 oersteds

								Pole position			
Site	N	n	$D°$	$I°$	R	K	$\delta°$	$W°$	$°S$	δ_p	δ_m
1	8	7	267	–04	6.987	462	9				
2*	9	8	259	–08	7.973	259	5				
4	6	6	258	18	5.877	41	11				
6*	5	5	256	–07	4.975	143	8				
7	6	6	237	27	5.679	17	15				
8*	6	3	252	–02	2.961	51	16				
9	6	6	258	23	5.942	86	6				
10*	20	18	255	–14	17.334	29	5				
12	7	5	207	34	4.719	14	20				
15	6	6	247	–07	5.917	60	9				
17	8	7	209	–02	6.824	31	19				
Mean	11		247	06	9.999	11	14	137	10	7.7	15.2

N = number of specimens measured.
n = number of specimens used.
R = resultant vector.
K = precision estimate.
$\delta°$ = angular standard deviation.
δ_p, δ_m = semi-axes of the ellipse of confidence at $P = 0.05$.
* = reversed polarization.

hand-samples is homogeneous after demagnetization. Two sites were removed from the analysis on the basis of inhomogeneity. The between-sample grouping of the data in any one site has shown relatively large inconsistencies on five of the remaining sites, subsequently only eleven of the original eighteen sites were retained in the analysis (Table 1).

9. Results of palaeomagnetic measurements

The N.R.M. directions of the Montagnais diabase dikes showed variable dispersion. Before AF demagnetization, the mean direction of the N.R.M. orientations in the diabase dikes is 289°, 22°S with a corresponding paleopole of 172°W, 12°N; after AF demagnetization between 100 and 200 oersteds (peak-to-peak), it becomes 247°, 06°N with a corresponding paleopole position of 137°W, 10°S ($K = 17$, $\alpha\, 0.95 = 14°$).

After AF demagnetization, the dispersion within and between sites decreased and the site mean directions grouped a north-east–south-west axis with low inclination; the dispersion of the poles still remained relatively large. There are seven sites with westerly declination (referred to as normal polarity) and four sites with easterly declination (reversed polarity).

The stable component remaining after AF cleaning is much smaller than the initial N.R.M. intensity; the average ratios of the mean site intensities, before and after cleaning at 200 oersteds, is 0.31. The palaeomagnetic significance and stability of the Montagnais diabases are demonstrated by the departure of the site N.R.M. directions from the recent earth field direction (332°, 78°N) and the evidence for reversal(s).

10. Curie temperature

Thermogravimetric curves were obtained on 10 mg of powdered-rock samples using a recording electromagnetic microbalance (Fig. 5). All the curves obtained are simple and indicate the presence of a single magnetic mineral having a Curie temperature in the 562°C to 590°C range with an average of 580°C. The data suggest that the magnetic mineral is almost pure magnetite. In spite of the fact that no titanomagnetite was detected, ilmenite was identified with a high resolution reflected light microscope.

11. Time (age) and origin of magnetization

Absolute age determinations, especially of Precambrian rocks, are difficult to obtain and their relationship to palaeomagnetic studies have been discussed by a number of authors (e.g. SPALL, 1971a, b; FAHRIG and LAROCHELLE, 1972; ROY and FAHRIG, 1973; PARK, IRVING and DONALDSON, 1973). The paleopole position of the Montagnais diabase dikes comes very close to the one of the Front Range intrusives of Colorado and Wyoming states (08°S, 151°W; 1410 m.y.), of the Michikamau anorthosite of Labrador (01°S, 145°W; 1400 m.y.), of the Sibley group sediments (20°S, 146°W; 1370 m.y.),

and of the Belt Series of Montana (12°S, 151°W; 1100–1500 m.y.). It is very unlikely that there has been any structural rotation of the area since the Labrador eugeosyncline, which makes part of the Churchill Province is lying in perfect depositional unconformity on the gneissic Ashuanipi Complex (age: 2500 m.y.) of the Superior Province to the west, which is itself the central nucleus of the Canadian Shield.

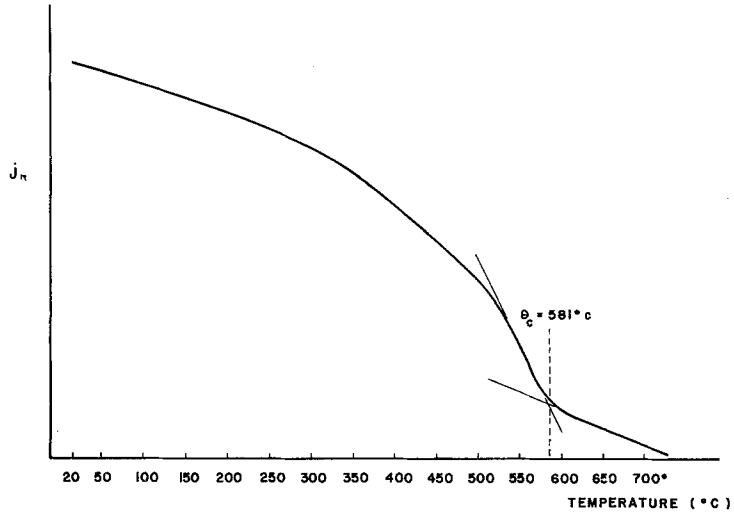


Figure 5
Thermogravimetric curve representative of the Montagnais diabase dikes (specimen No.: 72–8; magnetic field intensity: 800 oersteds)

The palaeomagnetic results suggest that the K–Ar minimum age of 1150 m.y. obtained from the Montagnais diabase dikes is too small and that the age of these dikes is most likely 1400 m.y.; this age would then correspond to the Elsonian orogeny (1370 m.y.). It is also observed that the Montagnais diabase dikes, like the Front Range intrusives, the Sibley Group and the Belt Series, depict mixed polarities; the Michikamau anorthosite, however, does not have any pole reversal. Mixed polarities in rocks of this age range between 1200 and 1400 m.y. (ROBERTSON, 1973).

12. Interpretation of paleopole, Helikian paleolatitude and climate

The pole position for the Montagnais diabase dikes falls in the South Pacific (close to Marquesas Islands, France), farther south than poles obtained from immediately older and younger rocks of the Canadian Shield. The inclination of the Montagnais rocks is low indicating that they were intruded in low latitude. During the Paleo-Helikian time (the oldest (lower) division of the middle Proterozoic Era), the central Labrador trough was situated well within the tropics.

13. Discussion of the results

The initial directions of magnetization are quite scattered, but the scattering is reduced by magnetic cleaning in a peak field of 200 oersteds. The scattering is attributed to temporary components acquired in the laboratory, viscous components acquired in the earth's magnetic field, a small fraction of chemical magnetization, secular variation, and measuring errors. A very high proportion of the opaque minerals are medium-sized magnetite crystals, which are considered to dominate the magnetic properties of the rocks. The pole position of the Montagnais dikes is consistent with the paleomagnetic data from North America rock units which have an age of 1300 to 1400 m.y., but the K–Ar minimum radiogenic age is only of the order of 1150 to 1200 m.y. Many sites yielded mixed polarities but the preponderance of normal polarities is significant. The reversed magnetization is thought to reflect a reversed ambient field at the time of intrusion. The paleolatitude at this time was close to the equator, suggesting a warm climate in Paleo-Helikian time.

No attempt has been made to discuss the path of apparent polar wandering related to the Canadian Shield during the Proterozoic Era, to suggest some modification to it, or to calculate the average rate of polar movement. In terms of tectonics, one can definitely say that the Montagnais diabase dikes have filled regional and local tension fractures of the earth's crust in Helikian Era. It is improbable that the state of regional tension present at that time was active enough to form a rift zone on a large scale.

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